

MOHAMED MOAMEN

Full-Stack Software Development Engineer

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PROFESSIONAL SUMMARY

Computer Science graduate and full-stack developer specializing in backend development and scalable web applications. with hands-on experience architecting complete systems end-to-end — from database design and REST APIs to React frontends and browser-level integrations. Comfortable working across multiple backend frameworks (Express, NestJS) and low-level APIs (WebAssembly, AudioWorklet), with coursework exposure to automated testing, CI/CD, and containerization, and eager to deepen these practices in a production environment focused on security and code quality.

TECHNICAL SKILLS

- **Full-Stack Development:** Node.js, Express.js, NestJS, TypeScript, JavaScript, React, REST APIs, GraphQL, MVC/Modular Architecture
- **Databases:** SQL, PostgreSQL, MySQL, SQLite, MongoDB
- **Security & Auth:** JWT Authentication, Token Rotation, bcrypt, Schema Validation (Zod), Secure Session Handling
- **DevOps & Testing (foundational):** Git, GitHub, CI/CD concepts, Docker fundamentals, unit/integration testing principles (coursework-based)
- **Additional:** Python, Flask, Django, Socket.IO, WebAssembly, Postman

PROJECTS

Gym Management System — Full-Stack Business Application ([GitHub](#))

- Designed a modular full-stack system covering member management, front-desk barcode check-ins, staff clock-in/out, subscription billing, and revenue analytics.
- Built the backend with NestJS using a feature-module architecture (auth, customers, check-ins, memberships, payments, staff, jobs), demonstrating scalable, maintainable service design.
- Implemented JWT authentication with forced password rotation on first login, and an hourly scheduled background job to automatically expire memberships.
- Integrated a third-party messaging API (OpenWA/WhatsApp) for automated member broadcasts, and documented a clear path to swap SQLite for PostgreSQL with TypeORM migrations for multi-server production deployment.

Real-Time Microphone Noise-Cancellation — Chrome Extension ([GitHub](#))

- Building a Manifest V3 Chrome Extension that performs real-time microphone noise cancellation using the AudioWorklet API and a WebAssembly-compiled RNNoise model.
- Working directly with low-level browser audio pipelines and cross-boundary WASM integration, and diagnosing OS/browser-level permission issues (microphone access errors) as part of the debugging process.

BizFlow — AI-Powered Business Planning Platform (Graduation Project) ([GitHub](#))

- Built a full-stack platform with a Node.js/Express backend and a React 19 + TypeScript + Vite frontend, integrating Google Gemini and Google Places/Maps APIs.
- Applied schema validation (Zod), secure authentication, and bilingual (Arabic/English) UI architecture across the stack.

Social Media App Backend ([GitHub](#))

- Built a TypeScript/Node.js/GraphQL backend with JWT + Redis-backed refresh-token rotation and Socket.IO real-time features.

EDUCATION

B.Sc. in Computer Science — Ahram Canadian University

2022 – 2026 | CGPA: Good

CERTIFICATIONS

- Backend Development (Node.js) — Route Academy, 6-Month Diploma
- Full Stack Web Development (Python) — Information Technology Institute (ITI), 120 hrs

LANGUAGES

- Arabic — Native | English — Good (Professional Working Proficiency)